

A small generator, a difficult observer

HMM/MSP primer · before Notebook 1 · 8–10 minutes

ILIAD Intensive · Computational Mechanics

What must an observer infer?

ZERO-ONE-RANDOM

The stream has a three-step rhythm. The phase is hidden.

0 1 1 0 1 0 0 1 1 ...

Generator's problem

Which hidden phase emits next, and where does it move?

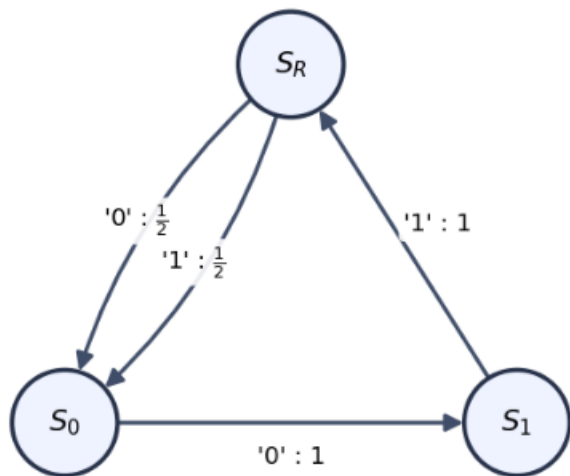
Observer's problem

Given only the symbols so far, what uncertainty should I retain?

A small generator does not imply a trivial prediction problem.

How to read one edge

HIDDEN GENERATOR — node = unseen phase; edge label = emitted symbol : probability



$$S_i \xrightarrow{x:p} S_j$$

means the **joint event**

$$\Pr (X_{t+1} = x, S_{t+1} = S_j \mid S_t = S_i) = p.$$

Timing convention: the state is the phase *before* the edge; traversing the edge emits a symbol and lands in the next phase.

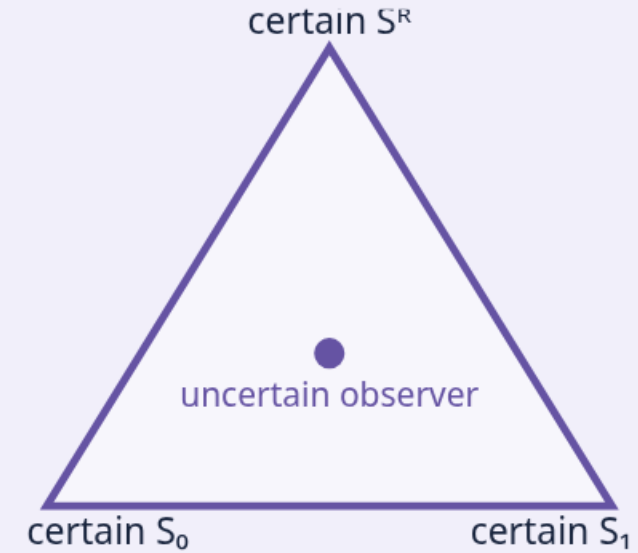
Generator state is not observer state

Hidden phase

One vertex of the generator diagram is physically occupied at each step.

This is part of a chosen realization of the process.

Observer belief



The notebook will turn uncertainty into a state that moves when symbols arrive.

Your derivation starts from the picture



OPEN NOTEBOOK 1

First task: reconstruct the symbol-labelled matrices from the diagram before any matrix is revealed.

- 1 Which hidden paths can emit a given word?
- 2 How should the observer update after each new symbol?
- 3 Can three hidden phases generate more than three observer states?